

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

FIRST SEMESTER M.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Mathematical Foundations for Data Science [MAT_5129]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Let X be Poisson(λ) random variable with probability mass function (3)
- A)
$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots; \lambda > 0.$$
- Find the MLE of λ .
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) A quality-control inspector takes a random sample of $n = 15$ bottles (4)
- from a filling machine. The sample mean is $\bar{x} = 24.7$ ml and the sample standard deviation is 0.40 ml. Test at the 5 % level whether the mean fill differs from the target mean 25 ml (use the test statistic value=2.145).
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) A sample of 20 observations had a sample standard deviation 37.2. Is (3)
- this compatible with the hypothesis that the sample is from a normal population with variance 4.35? (use the test statistic value=8.907)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) (4)
- A)

A coffee shop experiments with three roasting profiles (Light, Medium, Dark). For each profile the shop randomly selects 4 customers to rate the taste on a 1 to 10 scale. The scores are:

Customer	Roasting Profile		
	Light	Medium	Dark
1	7	6	8
2	8	5	7
3	6	7	9
4	9	6	8

Test whether mean taste scores differ by roasting profile, use $\alpha = 5\%$. (use the test statistic value=4.26).

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) List the three postulates of Poisson Process. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- C) A student alternates between four modes of study: (3)

$$S_1 = \text{Reading}, S_2 = \text{Watching Videos}, \\ S_3 = \text{Practicing Problems}, S_4 = \text{Taking Break}.$$

The following behavior is observed:

- a) From Reading: 50% continue reading, 30% switch to videos, 20% switch to practice.
- b) From videos: 40% move to practice, 40% remain in videos, 20% take a break.
- c) From Practice: 60% remain in practice, 20% go to reading, 20% go to break.
- d) From Break: 50% return to reading, 50% stay on break.

Write the transition probability matrix and draw the Markov chain diagram.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Define (in mathematical terms) transient, persistent, periodic and recurrent states of a Markov chain. (4)

- A) Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) Check whether the states of the Markov Chain are recurrent or transient. (3)

$S = \{1,2,3,4\}$ is the state space, and the transition probability matrix is :

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/4 & 3/4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Find the stationary distribution and comment on its existence for a Markov chain with transition matrix (3)

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$$

- 4) The following are five measurements on the variables X_1, X_2, X_3 . Find the mean vector μ and the covariance matrix Σ . (3)

A)

X_1	9	2	6	5	8
X_2	12	8	6	4	10
X_3	3	4	0	2	1

B)

If $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N_3(\mu, \Sigma)$. Write the formula for the mean vector and covariance matrix of $X' | X''$ where $X' = X_1$ and $X'' = \begin{bmatrix} X_2 \\ X_3 \end{bmatrix}$. (3)

- C) A random sample of size 3 chosen from a bivariate normal distribution be (4)

$X = \begin{bmatrix} 6 & 9 \\ 10 & 6 \\ 8 & 3 \end{bmatrix}$. Obtain the Hotelling's T^2 value for the population mean $\mu = \begin{bmatrix} 9 \\ 5 \end{bmatrix}$.

- 5) (3)

A)

Suppose the random variables X_1, X_2, X_3 have the covariance matrix

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \text{ The eigen values of } \Sigma \text{ matrix are } 5.83, 2, 0.17. \text{ The}$$

corresponding eigen vectors of these eigen values are

$$\begin{bmatrix} -0.38 \\ -0.92 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 0.92 \\ 0.38 \\ 0 \end{bmatrix}, \text{ respectively. Calculate}$$

- The three principal components
- Total variability of the principal components
- Correlation between first principal component and X_2 .

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) The covariance matrix of three random variables X_1, X_2, X_3 is given as (4)

$$\Sigma = \begin{bmatrix} 1 & 0.63 & 0.45 \\ 0.63 & 1 & 0.35 \\ 0.45 & 0.35 & 1 \end{bmatrix}. \text{ Assume } m = 1 \text{ factor model. Find}$$

- Matrix of factor loading (L) for the orthonormal factor model.
- Matrix of special variances (Ψ).
- Total communality of X_1, X_2, X_3 .

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) In PCA, $X_1, X_2, X_3, \dots, X_p$ be the original variables and $Z_1, Z_2, Z_3, \dots, Z_p$ (3)
be the principal components. What is the value of

- Variance of $Z_i, i \in \{1, 2, 3, \dots, p\}$
- $Cov(Z_i, Z_k), i, k \in \{1, 2, 3, \dots, p\}$
- Correlation coefficient $\rho_{(z_i, X_k)}, i, k \in \{1, 2, 3, \dots, p\}$

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM PG-2025-2026

MAT_5129:Mathematical Foundations for Data Science

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	17
3	3	CO 3	9
4	4	CO 4	7
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	7(14%)
3	Application	10	33(66%)
4	Analysis	3	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)